Introduction to Quantum Computing

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1 Basics concepts

Classical computing is based on the concept of bits, which can take values of either 0 or 1. **Quantum bits**, or **qubits**, on the other hand, can exist in a superposition of 0 and 1.

A quantum state $|\psi\rangle$ is described as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The ket notation of $|\psi\rangle$ stands for

$$|\psi\rangle \in \mathbb{C}^2, \quad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \implies |\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Difference from Classical Bit Unlike a classical bit, a qubit cannot be directly observed: the amplitudes α and β cannot be measured. A standard measurement will result in either 0 or 1, with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

Additionally, the measurement in general **changes** the qubit state ("wavefunction collapse"). For example, if we measure $|\psi\rangle$ and get 0, the state collapses to $|0\rangle$, and if we get 1, it collapses to $|1\rangle$.

In practise we can estimate the probabilities $|\alpha|^2$ and $|\beta|^2$ by repeating the same experiment many times (outcome statistics). These repetitions are called **shots** or **trials**.

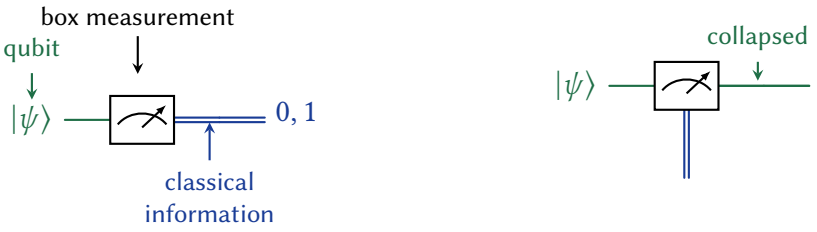


Figure 1: Circuit notation for a measurement.

What is a qubit physically? Many poossible realizations of qubits are>

- Two different polarizations of a photon (e.g.,left/right circular)
- Alignment of nuclear or electron spin ↑↓
- Ground state or excited state of an atom (electronic state)
- Clockwise or counterclockwise current in a superconducting loop with Josephson junctions (flux qubits)

Useful graphical representations of a qubit state are the **Bloch sphere**: if α and β happen to be real-valued that we can find angle $\theta \in \mathbb{R}$ such that $\alpha = \cos(\theta/2)$ and $\beta = \sin(\theta/2)$

$$|\alpha|^2 + |\beta|^2 = \cos^2(\theta/2) + \sin^2(\theta/2) = 1.$$

In general, α and β are complex numbers. Using so-called phase angles γ for α and $\gamma + \varphi$ for β, we represent

$$\alpha = e^{i\gamma}\cos(\theta/2), \qquad \beta = e^{i(\gamma+\varphi)}\sin(\theta/2), \quad \theta \in [0, \pi], \ \varphi \in [0, 2\pi).$$

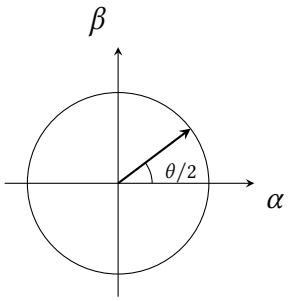


Figure 2: Real-valued α, β: the state is a point on the unit circle.

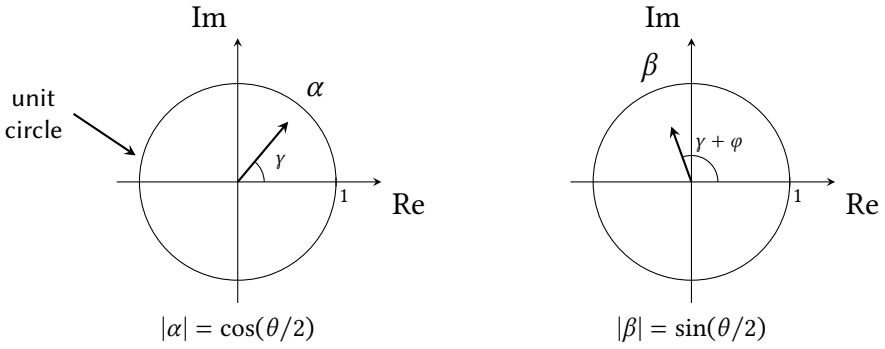


Figure 3: Complex planes of α and β.

Then

$$\begin{aligned} |\psi\rangle &= e^{i\gamma}\cos(\theta/2)|0\rangle + e^{i(\gamma+\varphi)}\sin(\theta/2)|1\rangle \\ &= e^{i\gamma}\left(\cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle\right) \end{aligned}$$

thus, up to the global phase $e^{i\gamma}$ (which is not observable), $|\psi\rangle$ is characterized by two angles θ and φ and these specify the point on the **Bloch sphere** defined as:

$$\vec{r} = \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} \in \mathbb{R}^3$$

On the surface of a sphere:

2 Single-qubit gates

2.1 Time evolution and circuit notation

Principle of time evolution The quantum state evolves over time. A quantum state $|\psi\rangle$ at a current time t transitions to a new quantum state $|\psi'\rangle$ at a later time point $t' > t$. The transition is described by a complex unitary matrix $\mathcal{U} \in \mathbb{C}^{2\times 2}$:

$$|\psi'\rangle = \mathcal{U}|\psi\rangle$$

a matrix vector multiplication.

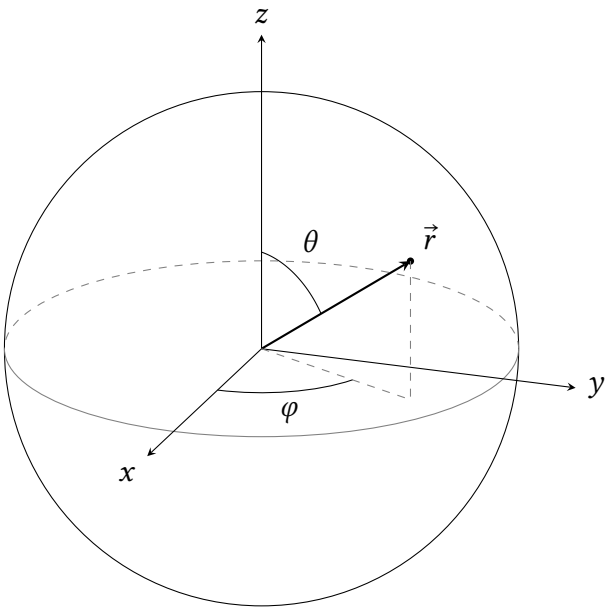


Figure 4: Bloch sphere: the point $\vec{r}$ is specified by the angles θ and φ.

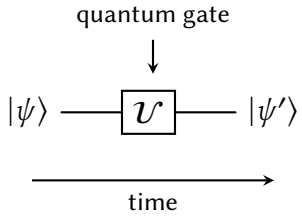


Figure 5: Circuit notation for a single-qubit gate.

Circuit Notation The circuit is read from left to right, but matrix times vector multiplication is from right to left ($\mathcal{U}|\psi\rangle$). Additionally, $\mathcal{U}$ preserves normalization: since $\mathcal{U}$ is unitary ($\mathcal{U}^\dagger\mathcal{U} = \mathbb{I}$), $\|\mathcal{U}|\psi\rangle\|^2 = \langle\psi \,|\, \mathcal{U}^\dagger\mathcal{U} \,|\, \psi\rangle = \langle\psi \,|\, \psi\rangle = 1$.

2.2 Common gates

Example 2.1 (A quantum analog of a classical NOT gate)**.** The classical NOT gate flips a bit $0 \mapsto 1$ and $1 \mapsto 0$. The quantum analog of a NOT gate is the **Pauli-X gate**, which flips the quantum state $|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$. The Pauli-X gate is represented by

$$X \equiv \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle,$$
$$X|1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

In addition we define the **Pauli-Y gate**

$$Y \equiv \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

and the **Pauli-Z gate**

$$Z \equiv \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Z leaves $|0\rangle$ unchanged, but flips the sign of the coefficient of $|1\rangle$. We recall the **Bloch sphere representation**

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle,$$

then

$$\begin{aligned} Z|\psi\rangle &= \cos(\theta/2)|0\rangle - e^{i\varphi}\sin(\theta/2)|1\rangle \\ &= \cos(\theta/2)|0\rangle - \underbrace{e^{i\pi}e^{i\varphi}}_{e^{i(\pi+\varphi)}}\sin(\theta/2)|1\rangle \end{aligned}$$

remembering that $e^{i\pi} = -1$. New bloch sphere coordinates are $\theta' = \theta$ and $\varphi' = \pi + \varphi$, so the Z gate corresponds to a rotation of the Bloch sphere around the z-axis by π.

Proof that $e^{i\pi} = -1$. Recall the power series, valid for all $z \in \mathbb{C}$:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}, \qquad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}.$$

Let $x \in \mathbb{R}$ and set $z = ix$. Since $i^{2k} = (-1)^k$ and $i^{2k+1} = i\,(-1)^k$, splitting the series into even and odd terms (allowed by absolute convergence) gives

$$e^{ix} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = \cos x + i \sin x.$$

Evaluating at $x = \pi$, where $\cos \pi = -1$ and $\sin \pi = 0$, we obtain

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0 = -1. \qquad \square$$

X, Y, Z gates are also called **Pauli matrices** and are Hermitian, i.e. $X^\dagger = X$, $Y^\dagger = Y$, $Z^\dagger = Z$. They are also unitary, i.e. $X^{-1} = X^\dagger$, $Y^{-1} = Y^\dagger$, $Z^{-1} = Z^\dagger$.

The **Pauli Vector** is defined as $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3) = (X, Y, Z)$ is a vector of 2×2 matrices.

Hadamard Gate We define the **Hadamard gate** H as

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

so that

$$\begin{aligned} H|\psi\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{\sqrt{2}}(\alpha + \beta) \\ \frac{1}{\sqrt{2}}(\alpha - \beta) \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle \\ &= \alpha \frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}$$

$$\alpha|0\rangle + \beta|1\rangle \quad \boxed{H} \quad \alpha\,\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \beta\,\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Figure 6: Hadamard gate on a general state.

Phase Gate: S and T We define the **phase gate** S as

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \implies S^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

and the **phase gate** T as

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \implies T^2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} = S$$

since $(e^{i\pi/4})^2 = e^{i\pi/2} = i$.

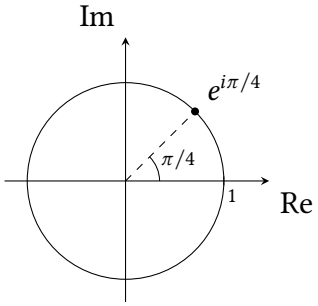


Figure 7: Phase of the T gate.

Remark. The Pauli matrices satisfy the following relations. Since each σ_j is Hermitian and unitary,

$$\sigma_j^2 = \sigma_j \sigma_j^\dagger = \mathbb{I} \quad \text{for } j = 1, 2, 3, \quad \text{i.e. } X^2 = Y^2 = Z^2 = \mathbb{I}.$$

Additionally, the Pauli matrices anticommute:

$$\sigma_j \sigma_k = -\sigma_k \sigma_j, \quad \forall j \neq k$$

We define the commutator of two matrices A and B as

$$[A, B] := AB - BA$$

and for the Pauli matrices we have

$$[\sigma_j, \sigma_k] = \sigma_j \sigma_k - \sigma_k \sigma_j = 2i\sigma_l$$

for (j, k, l) a cyclic permutation of $(1, 2, 3)$ e.g. $j = 2, k = 3, l = 1$. So for example,

$$[\sigma_2, \sigma_3] = 2i\sigma_1$$
$$[Y, Z] = 2iX$$

where the cyclic permutations of $(1, 2, 3)$ are $(1, 2, 3)$, $(2, 3, 1)$, $(3, 1, 2)$.

2.3 The matrix exponential

General definition of the **matrix exponential** (via power series expansion):

$$e^A \equiv \exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k, \quad A \in \mathbb{C}^{n\times n}$$

with $A^k = A \cdot A \cdots A$ (k times) and $n \in \mathbb{N}$.

Generalization of Euler's formula Special case: self-inverse matrix $A^2 = \mathbb{I}$ (identity), and $x \in \mathbb{R}$ a real number. Then

$$e^{iAx} = \cos(x)\mathbb{I} + i\sin(x)A,$$

which generalizes Euler's formula $e^{ix} = \cos(x) + i\sin(x)$.

Proof. Since $A^2 = \mathbb{I}$, it is clear that $A^k = \begin{cases} \mathbb{I} & k \text{ even} \\ A & k \text{ odd} \end{cases}$. Therefore, the power series expansion of the exponential can be partitioned as

$$\begin{aligned} e^{iAx} &= \mathbb{I} + iAx - \frac{1}{2!}\mathbb{I}x^2 - \frac{i}{3!}Ax^3 + \ldots \\ &= \mathbb{I}\left(1 - \frac{1}{2!}x^2 + \ldots\right) + iA\left(x - \frac{1}{3!}x^3 + \ldots\right). \end{aligned}$$

Note that the series inside the brackets correspond to the Taylor expansions of cos and sin:

$$\begin{aligned} \cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots \end{aligned}$$

In summary,

$$e^{iAx} = \cos(x)\mathbb{I} + i\sin(x)A.$$

□

Unitary matrix exponentials If $A \in \mathbb{C}^{n\times n}$ ($n \in \mathbb{N}$) is Hermitian, then e^{iA} is unitary.

A is called **Hermitian** if $A^\dagger = A$, with $A^\dagger = (A^*)^T$ denoting the conjugate transpose. $U \in \mathbb{C}^{n\times n}$ is called **unitary** if $U^\dagger U = \mathbb{I}$.

Proof. We first observe that

$$\left(e^{iA}\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left((iA)^k\right)^\dagger = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-iA^\dagger\right)^k = \sum_{k=0}^{\infty} \frac{1}{k!}(-iA)^k = e^{-iA}.$$

For any square matrices A and B which *commute* ($[A, B] = AB - BA = 0$), it holds that $e^A e^B = e^{A+B}$. (This can be shown analogous to the scalar case, but does not hold in general if $[A, B] \neq 0$.) Thus

$$\left(e^{iA}\right)^\dagger e^{iA} = e^{-iA} e^{iA} = e^{-iA+iA} = e^0 = \mathbb{I}.$$

Here we have used that $-iA$ commutes with iA (since these matrices only differ by a sign factor). □

Single-qubit rotation operators Pauli matrices X, Y, Z are Hermitian and self-inverse. Can be used to define the following unitary *rotation operators*, with parameter $\theta \in \mathbb{R}$:

$$R_x(\theta) := e^{-i\theta X/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)X = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_y(\theta) := e^{-i\theta Y/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)Y = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_z(\theta) := e^{-i\theta Z/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)Z = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

General case: rotation about an axis $\vec{v} \in \mathbb{R}^3$ (normalized such that with $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} = 1$):

Notation

$$\underbrace{\vec{v} \cdot \vec{\sigma}}_{\langle \vec{v} | \vec{\sigma} \rangle} \equiv v_1 X + v_2 Y + v_3 Z = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix} \in \mathbb{C}^{2\times 2}$$

It holds that $\vec{v} \cdot \vec{\sigma}$ is Hermitian and $(\vec{v} \cdot \vec{\sigma})^2 = \mathbb{I}$:

$$\begin{aligned}(\vec{v} \cdot \vec{\sigma})^2 &= \begin{pmatrix} v_3^2 + (v_1 - iv_2)(v_1 + iv_2) & v_3(v_1 - iv_2) + (v_1 - iv_2)(-v_3) \\ (v_1 + iv_2)v_3 - v_3(v_1 + iv_2) & v_3^2 + (v_1 - iv_2)(v_1 + iv_2) \end{pmatrix} \\ &= \begin{pmatrix} v_3^2 + v_1^2 + v_2^2 & 0 \\ 0 & v_3^2 + v_1^2 + v_2^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

where in the last step we have used that $\vec{v}$ is a unit vector, and therefore $\|\vec{v}\|^2 = v_1^2 + v_2^2 + v_3^2 = 1$.

General unitary *rotation operator about axis* $\vec{v}$ with angle $\theta \in \mathbb{R}$:

$$R_{\vec{v}}(\theta) := e^{-i\theta(\vec{v}\cdot\vec{\sigma})/2} = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)(\vec{v} \cdot \vec{\sigma}).$$

Note: R_x, R_y, R_z are special cases, corresponding to $\vec{v} = (1, 0, 0)$, $\vec{v} = (0, 1, 0)$, $\vec{v} = (0, 0, 1)$.

Can derive that Bloch sphere representation of $R_{\vec{v}}(\theta)$ is a "conventional" rotation (in three dimensions) by angle θ about axis $\vec{v}$ (see Tutorial 11): with $\vec{r}$ the Bloch vector of $|\psi\rangle$,

$$|\psi'\rangle = R_{\vec{v}}(\theta)|\psi\rangle$$

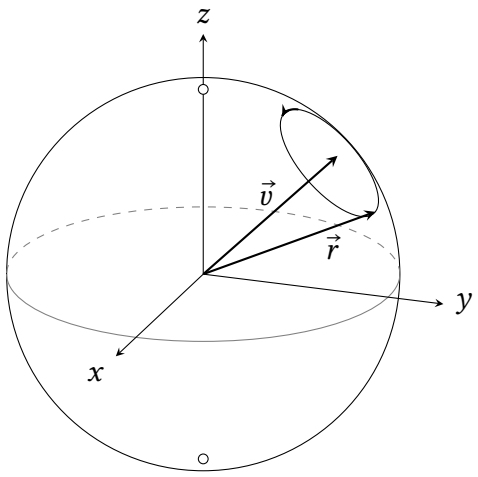


Figure 8: Rotation of the Bloch vector $\vec{r}$ about the axis $\vec{v}$.

3 Multiple qubits

So far we have only considered a single qubit, superpositions of basis states $|0\rangle$ and $|1\rangle$. For two qubits, this generalizes to

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

as computational basis states: all combinations (bit strings) of 0s and 1s. The generic two-qubit state is a superposition of these four basis states:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle,$$

with amplitudes $\alpha_{ij} \in \mathbb{C}$, $i, j \in \{0, 1\}$, such that

$$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$$

We can identify the basis states with vectors in $\mathbb{C}^4$:

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Thus, a two-qubit state can be represented as:

$$|\psi\rangle = \begin{pmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{pmatrix} \in \mathbb{C}^4$$

See what happens if we measure only one qubit of a two-qubit state. For example, we measure the first qubit and we obtain the result

$$\begin{aligned} &0 \text{ with probability } |\alpha_{00}|^2 + |\alpha_{01}|^2 \\ &1 \text{ with probability } |\alpha_{10}|^2 + |\alpha_{11}|^2 \end{aligned}$$

Wavefunction directly after measurement

$$\begin{aligned} \text{if measurement is } 0 \ : \quad |\psi'\rangle &= \frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}} \\ \text{if measurement is } 1 \ : \quad |\psi'\rangle &= \frac{\alpha_{10}|10\rangle + \alpha_{11}|11\rangle}{\sqrt{|\alpha_{10}|^2 + |\alpha_{11}|^2}} \end{aligned}$$

where the denominator is the normalization factor. We now introduce the mathematical tool to construct two-qubit states: the tensor product of vector spaces.

3.1 Tensor products of vector spaces

Can combine two (arbitrary) vector spaces V and W to form the **tensor product** $V \otimes W$. The elements of $V \otimes W$ are linear combinations of "tensor products" $|v\rangle \otimes |w\rangle$ consisting of elements $|v\rangle \in V$ and $|w\rangle \in W$.

Example 3.1. Let $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ be the single qubit spaces with basis $\{|0\rangle, |1\rangle\}$, then

$$\frac{1}{2} \underbrace{|0\rangle \otimes |0\rangle}_{=|00\rangle} + \frac{5i}{7} \underbrace{|1\rangle \otimes |0\rangle}_{=|10\rangle}$$

is an element of $V \otimes W$.

Let $\{|i\rangle_V : i = 1, \dots, m\}$ be a basis of V, and $\{|j\rangle_W : j = 1, \dots, n\}$ be a basis of W, then

$$\{|i\rangle_V \otimes |j\rangle_W : i = 1, \dots, m,\ j = 1, \dots, n\}$$

is a basis of $V \otimes W$. In particular:

$$\dim(V \otimes W) = \dim(V) \cdot \dim(W).$$

Note: $|i\rangle_V \otimes |j\rangle_W$ is also written as $|i\rangle|j\rangle$ or $|ij\rangle$.

Example 3.2. The basis construction for $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ leads to the standard two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

Elementary properties of tensor products:

- for all $|v\rangle \in V$, $|w\rangle \in W$ and $\alpha \in \mathbb{C}$:

$$\alpha(|v\rangle \otimes |w\rangle) = (\alpha|v\rangle) \otimes |w\rangle = |v\rangle \otimes (\alpha|w\rangle)$$

- for all $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$:

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle$$

- for all $|v\rangle \in V$ and $|w_1\rangle, |w_2\rangle \in W$:

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle$$

Vector notation (using standard basis), e.g.,

$$|v\rangle = v_1|0\rangle + v_2|1\rangle = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad |w\rangle = w_1|0\rangle + w_2|1\rangle = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix},$$

then:

$$\begin{aligned} |v\rangle \otimes |w\rangle &= (v_1|0\rangle + v_2|1\rangle) \otimes (w_1|0\rangle + w_2|1\rangle) \\ &= v_1 w_1|00\rangle + v_1 w_2|01\rangle + v_2 w_1|10\rangle + v_2 w_2|11\rangle \end{aligned}$$

Thus:

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{pmatrix}$$

Example 3.3.

$$\begin{pmatrix}2\\3\end{pmatrix}\otimes\begin{pmatrix}5\\7\end{pmatrix}=\begin{pmatrix}10\\14\\15\\21\end{pmatrix}$$

Note: not every element of $V\otimes W$ can be written in the form $|v\rangle\otimes|w\rangle$ with $|v\rangle\in V$ and $|w\rangle\in W$, for example the so-called **Bell state**

$$|\psi\rangle=\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle).$$

Assuming that V and W are equipped with an inner (scalar) product $\langle\cdot\,|\,\cdot\rangle$, one can define an inner product on $V\otimes W$ by:

$$\Big\langle\sum_j\alpha_j|v_j\rangle|w_j\rangle\;\Big|\;\sum_k\beta_k|\tilde{v}_k\rangle|\tilde{w}_k\rangle\Big\rangle:=\sum_j\sum_k\alpha_j^*\beta_k\langle v_j\,|\,\tilde{v}_k\rangle\langle w_j\,|\,\tilde{w}_k\rangle.$$

3.2 Generalization to n-qubits: 2^n computational basis states

$$\{\underbrace{|0,\dots,0\rangle}_{\text{length }n},|0,\dots,1\rangle,\dots,|1,\dots,1\rangle\}$$

Thus, a general n-qubit quantum state also denoted as "Quantum Register" is given by:

$$|\psi\rangle=\sum_{x_0=0}^{1}\sum_{x_1=0}^{1}\cdots\sum_{x_{n-1}=0}^{1}\alpha_{x_0,x_1,\dots,x_{n-1}}|x_0,x_1,\dots,x_{n-1}\rangle=\sum_{x=0}^{2^n-1}\alpha_x|x\rangle$$

where x is the integer with binary digits $x_0x_1\dots x_{n-1}$, and $\alpha_x\in\mathbb{C}$ for all $x\in\{0,1,\dots,2^n-1\}$ such that

$$\||\psi\rangle\|^2=\sum_{x=0}^{2^n-1}|\alpha_x|^2=1\quad\text{(normalization)}$$

In general, since the number 2^n of amplitudes grows exponentially with n, this is "hard" to simulate on a classical computer (for large n). This effect is known as the "curse of dimensionality".

We can see the vector space as a tensor product:

$$\underbrace{\mathbb{C}^2\otimes\mathbb{C}^2\otimes\dots\otimes\mathbb{C}^2}_{n\text{ times}}=(\mathbb{C}^2)^{\otimes n}=\mathbb{C}^{2^n}$$

3.3 Multi-qubit gates

As for single-qubit, an operation on multiple qubits is described by a unitary matrix $\mathcal{U}$. For n qubits, $\mathcal{U}\in\mathbb{C}^{2^n\times 2^n}$.

Example 3.4 (Controlled-NOT (CNOT) gate)**.** The CNOT gate is a two-qubit gate and it has a control qubit and a target qubit. The target qubit is flipped if the control qubit is in state $|1\rangle$, otherwise it is left unchanged.

$$|00\rangle\mapsto|00\rangle,\quad|01\rangle\mapsto|01\rangle,\quad|10\rangle\mapsto|11\rangle,\quad|11\rangle\mapsto|10\rangle$$

with the control qubit (first, in red) and the target qubit (second).

It can be expressed as:

$$|a, b\rangle \mapsto |a, a \oplus b\rangle \quad \text{for } a, b \in \{0, 1\}$$

where $\oplus$ denotes addition modulo 2.

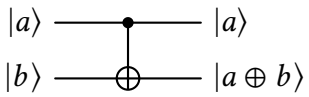


Figure 9: Circuit notation for the CNOT gate.

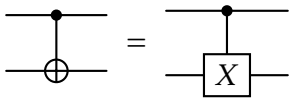
Matrix representation (in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$):

$$\mathcal{U}_{\text{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Pauli-X

$\mathcal{U}_{\text{CNOT}}$ is unitary: it is real and symmetric, and $\mathcal{U}_{\text{CNOT}}^2 = \mathbb{I}$ (applying it twice flips the target back), so $\mathcal{U}_{\text{CNOT}}^\dagger \mathcal{U}_{\text{CNOT}} = \mathcal{U}_{\text{CNOT}}^2 = \mathbb{I}$.

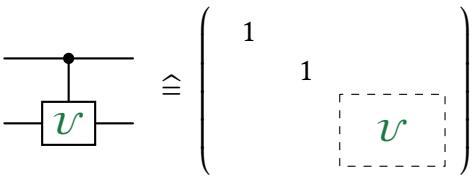
Alternative circuit notation:



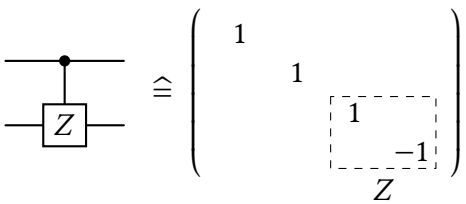
Controlled-$\mathcal{U}$ gates We can generalize Pauli-X to any unitary operator $\mathcal{U} \in \mathbb{C}^{2\times 2}$ acting on the target qubit, which gives the **controlled-$\mathcal{U}$ gate**:

$$|00\rangle \mapsto |00\rangle, \quad |01\rangle \mapsto |01\rangle, \quad |10\rangle \mapsto |1\rangle \otimes (\mathcal{U}|0\rangle), \quad |11\rangle \mapsto |1\rangle \otimes (\mathcal{U}|1\rangle).$$

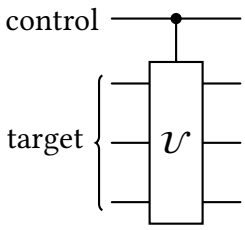
In circuit and matrix form:



Example 3.5 (Controlled-Z gate)**.** For $\mathcal{U} = Z$:



Controlled-$\mathcal{U}$ for multiple target qubits The control qubit can also act on several target qubits, with $\mathcal{U}$ a unitary on all of them:



Remark. Single-qubit and CNOT gates are **universal**: they can be used to implement an arbitrary unitary operation on n qubits (quantum analog of the universality of the classical NAND gate).

Example 3.6 (A circuit of single-qubit gates and CNOTs)**.** A circuit on three qubits consisting only of single-qubit gates and CNOTs:

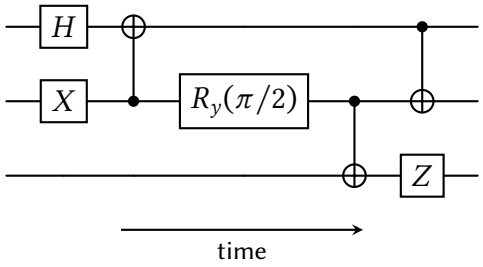
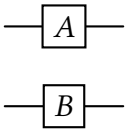


Figure 10: A circuit consisting only of single-qubit gates and CNOTs.

Kronecker products Matrix representation of single-qubit gates acting in parallel:



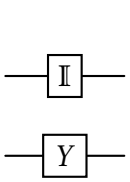
On the basis states $|a, b\rangle = |a\rangle \otimes |b\rangle$, $a, b \in \{0, 1\}$:

$$|a, b\rangle \mapsto (A|a\rangle) \otimes (B|b\rangle) = (A \otimes B)|a, b\rangle.$$

Example 3.7 ($A = \mathbb{I}$, $B = Y$)**.**

$$\begin{aligned}
|00\rangle &\mapsto |0\rangle \otimes (\underbrace{Y|0\rangle}_{i|1\rangle}) = i|01\rangle, \\
|01\rangle &\mapsto |0\rangle \otimes (\underbrace{Y|1\rangle}_{-i|0\rangle}) = -i|00\rangle, \\
|10\rangle &\mapsto |1\rangle \otimes (Y|0\rangle) = i|11\rangle, \\
|11\rangle &\mapsto |1\rangle \otimes (Y|1\rangle) = -i|10\rangle.
\end{aligned}$$

Matrix representation (the images of the basis states are the columns):



$$\widehat{=} \begin{pmatrix} \overbrace{\begin{matrix} 0 & -i \\ i & 0 \end{matrix}}^{Y} & \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} & \overbrace{\begin{matrix} 0 & -i \\ i & 0 \end{matrix}}^{Y} \end{pmatrix} = \begin{pmatrix} Y & 0_{2\times 2} \\ 0_{2\times 2} & Y \end{pmatrix} = \mathbb{I} \otimes Y.$$

In general, the matrix representation of the tensor product of operators is the **Kronecker product**: for $A \in \mathbb{C}^{m\times n}$, $B \in \mathbb{C}^{p\times q}$,

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix} \in \mathbb{C}^{mp\times nq}.$$

In NumPy: `np.kron(A, B)`.